Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in DOE-guided process development, physical/organic chemistry, analytical characterization, and stability analysis across polymer, thin-film, and functional molecular systems. Builds reproducible process windows, material-quality readouts, SOP-ready documentation, and technical reports that support formulation, scale-up, and manufacturing decisions.

CORE COMPETENCIES

Oral Formulation Sciences | DOE / Process Development | Physical & Organic Chemistry | Materials Science | Scale-Up Documentation | GMP Readiness | Stability Analysis | Technical Reports

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided process-development experiments for bio-based polymer systems, defining process windows, material-quality readouts, reproducibility criteria, and SOP-ready specifications for scale-up.
- Use FT-IR, thermal analysis, spectroscopy, and SPC concepts to connect process parameters with material stability, robustness, and method repeatability.
- Translate process and characterization data into technical reports, troubleshooting recommendations, and documented methods suitable for formulation/process review.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Developed and characterized 15+ organic functional-material variants, connecting molecular structure, solid-state behavior, redox stability, and optical performance through multi-technique analysis.
- Executed synthetic routes, purification, NMR/MS/GPC-informed analysis, UV-Vis-NIR spectroscopy, cyclic voltammetry, and thermal characterization to diagnose stability and performance limits.
- Produced publication-quality figures, experimental records, and technical narratives that translate complex chemistry data into decision-ready conclusions.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded process development for molecular thin-film coatings, optimizing surface preparation and CVD conditions to improve reproducibility and optical/material performance.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition under flow using DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and process controls.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30% and strengthened data integrity.

Georgia Tech 2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

DOE-Guided Polymer Process Development Current

Defined process windows, analytical readouts, reproducibility criteria, and SOP-ready specifications for bio-based polymer scale-up work, mirroring formulation/process robustness needs.

DOE, SPC, FT-IR, thermal analysis, process variables, SOPs

Formulation-Adjacent Stability and Materials Analysis Chemistry of Materials 2025

Mapped degradation, redox-instability, and performance failure modes across functional material variants using orthogonal analytical characterization and physical/organic chemistry reasoning.

UV-Vis-NIR, FT-IR, CV, TGA/DSC, NMR, GPC, MS-informed analysis

AFOSR Thin-Film Coating Process Development DoD Funded

Connected CVD and surface-preparation variables to coating morphology, interfaces, and device-level performance, building evidence for process optimization and scale-up decisions.

CVD, QCM-D, spectroscopy, process documentation, coating robustness

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Process & Reaction Development: DOE, process characterization, route/reaction analysis, robustness/productivity studies, scale-up specifications, SOPs, QbD/cGMP/GDP readiness
Analytical Characterization: NMR, LC-MS exposure, GC-MS, GPC/SEC, FT-IR, UV-Vis-NIR, TGA, DSC, QCM-D, cyclic voltammetry, chromatography-adjacent workflows
Chemistry & Materials: Organic synthesis, purification, crystallization concepts, polymers, small-molecule materials, structure-property-process analysis
Data & Documentation: SPC, Excel, Python/basic, JMP/basic, electronic/structured lab records, technical reports, publication-quality writing

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)